10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

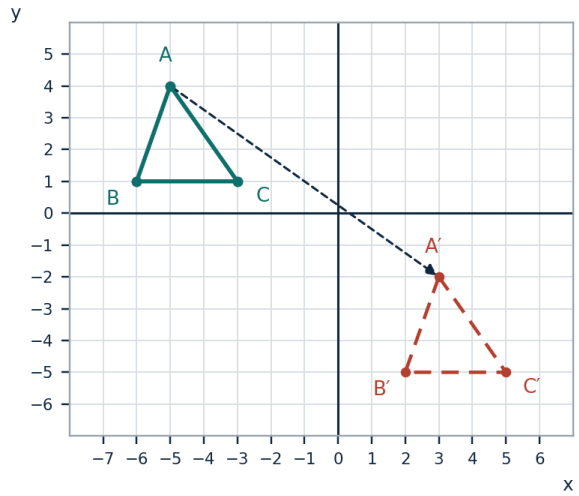
Exercise 6.3 — Translation and Geometric Problems

Unit 6 | Solved Exercise | New Book (PECTAA)

Exercise 6.3 (Solved)

A translation by the vector ai + bj moves every point (x, y) to (x + a, y + b): slide a units along x and b units along y. Lengths and angles do not change, so the image is congruent to the original figure.

1. Plot A(−5, 4), B(−6, 1) and C(−3, 1) to form a triangle ABC. Also translate ABC to A′B′C′ by the translation vector 8i − 6j.



ABC → A′B′C′ (8i − 6j)

For 8i − 6j add 8 to x, −6 to y

A′ = (−5 + 8)i + (4 − 6)j

= 3i − 2j = (3, −2)

B′ = (−6 + 8)i + (1 − 6)j

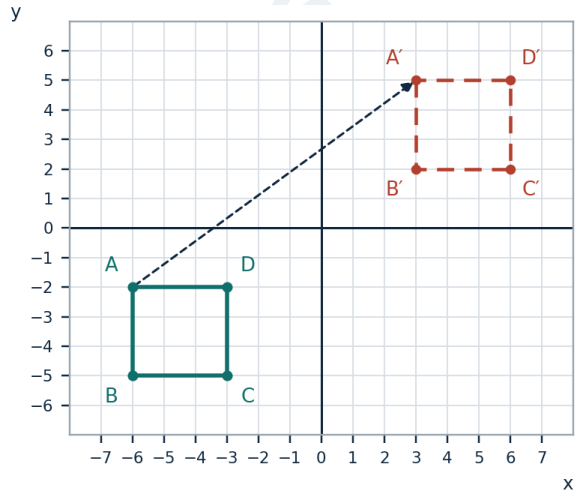
= 2i − 5j = (2, −5)

C′ = (−3 + 8)i + (1 − 6)j

= 5i − 5j = (5, −5)

Answer: A′(3, −2), B′(2, −5), C′(5, −5)

2. Plot A(−6, −2), B(−6, −5), C(−3, −5) and D(−3, −2) to form a square ABCD. Also translate square ABCD to square A′B′C′D′ by the translation vector 9i + 7j.



Square ABCD → A′B′C′D′ (9i + 7j)

For 9i + 7j add 9 to x, 7 to y

A′ = (−6 + 9)i + (−2 + 7)j = (3, 5)

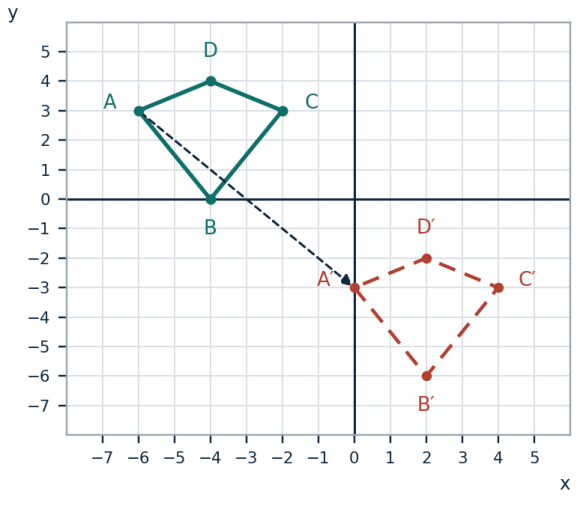
B′ = (−6 + 9)i + (−5 + 7)j = (3, 2)

C′ = (−3 + 9)i + (−5 + 7)j = (6, 2)

D′ = (−3 + 9)i + (−2 + 7)j = (6, 5)

Answer: A′(3, 5), B′(3, 2), C′(6, 2), D′(6, 5)

3. Plot A(−6, 3), B(−4, 0), C(−2, 3) and D(−4, 4) to form a kite ABCD. Also translate kite ABCD to kite A′B′C′D′ by the translation vector 6i − 6j.



Kite ABCD → A′B′C′D′ (6i − 6j)

For 6i − 6j add 6 to x, −6 to y

A′ = (−6 + 6)i + (3 − 6)j

= 0i − 3j = (0, −3)

B′ = (−4 + 6)i + (0 − 6)j

= 2i − 6j = (2, −6)

C′ = (−2 + 6)i + (3 − 6)j

= 4i − 3j = (4, −3)

D′ = (−4 + 6)i + (4 − 6)j

= 2i − 2j = (2, −2)

Answer: A′(0, −3), B′(2, −6), C′(4, −3), D′(2, −2)

4. The coordinates of A, B and D are (1, 2), (6, 3) and (2, 8) respectively. Find the coordinates of C by using the vector method if ABCD is a parallelogram.

$\overline{AB} = (6 - 1)\bar{i} + (3 - 2)\bar{j} = 5\bar{i} + \bar{j}$

In a parallelogram the opposite sides are equal and parallel, so $\overline{DC} = \overline{AB}$.

$\overline{DC} = C - D \implies C = D + \overline{AB}$

$C = (2\bar{i} + 8\bar{j}) + (5\bar{i} + \bar{j})$

$= (2 + 5)\bar{i} + (8 + 1)\bar{j}$

$= 7\bar{i} + 9\bar{j}$

Answer: C = (7, 9)

5. In parallelogram ABCD, the vectors representing two opposite sides are AB = 6i + 2j and DC = −6i − 2j. Show that the opposite sides are equal in magnitude and parallel.

$|\overline{AB}| = \sqrt{((6)^2 + (2)^2)} = \sqrt{(36 + 4)} = \sqrt{40}$

$|\overline{DC}| = \sqrt{((-6)^2 + (-2)^2)} = \sqrt{(36 + 4)} = \sqrt{40}$

So $|\overline{AB}| = |\overline{DC}| = 2\sqrt{10}$

Also $\overline{DC} = -(6\bar{i} + 2\bar{j}) = -\overline{AB}$

One vector is a scalar multiple of the other, so AB and DC are parallel.

Answer: |AB| = |DC| = 2√10 and AB ∥ DC

6. Points A(1, 2), B(4, 6) and C(7, 2) form a triangle. Check whether triangle ABC is an isosceles by using vector magnitude.

$\overline{AB} = (4 - 1)\bar{i} + (6 - 2)\bar{j} = 3\bar{i} + 4\bar{j}$

$\overline{BC} = (7 - 4)\bar{i} + (2 - 6)\bar{j} = 3\bar{i} - 4\bar{j}$

$\overline{AC} = (7 - 1)\bar{i} + (2 - 2)\bar{j} = 6\bar{i}$

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$|\overline{AB}| = \sqrt{(9 + 16)} = \sqrt{25} = 5$

$|\overline{BC}| = \sqrt{(9 + 16)} = \sqrt{25} = 5$

$|\overline{AC}| = \sqrt{(36 + 0)} = 6$

Two sides are equal in length, so the triangle is isosceles.

Answer: Yes, ΔABC is isosceles because |AB| = |BC| = 5

7. Use vectors to show that PQRS is a parallelogram, where the points P, Q, R and S have coordinates (1, 2), (5, 2), (7, 6) and (3, 6) respectively.

$\overline{PQ} = (5 - 1)\bar{i} + (2 - 2)\bar{j} = 4\bar{i}$

$\overline{SR} = (7 - 3)\bar{i} + (6 - 6)\bar{j} = 4\bar{i}$

$\overline{QR} = (7 - 5)\bar{i} + (6 - 2)\bar{j} = 2\bar{i} + 4\bar{j}$

$\overline{PS} = (3 - 1)\bar{i} + (6 - 2)\bar{j} = 2\bar{i} + 4\bar{j}$

PQ = SR and QR = PS, so both pairs of opposite sides are equal and parallel.

Answer: PQRS is a parallelogram

8. Use vectors to show that triangle XYZ is an isosceles, where the points X, Y and Z have the coordinates (0, 0), (2, 0) and (1, 3) respectively.

$\overline{XY} = (2 - 0)\bar{i} + (0 - 0)\bar{j} = 2\bar{i}$

$\overline{YZ} = (1 - 2)\bar{i} + (3 - 0)\bar{j} = -\bar{i} + 3\bar{j}$

$\overline{ZX} = (0 - 1)\bar{i} + (0 - 3)\bar{j} = -\bar{i} - 3\bar{j}$

$|\overline{XY}| = \sqrt{(4 + 0)} = 2$

$|\overline{YZ}| = \sqrt{(1 + 9)} = \sqrt{10}$

$|\overline{ZX}| = \sqrt{(1 + 9)} = \sqrt{10}$

YZ and ZX are equal in length, so the triangle is isosceles.

Answer: Yes, |YZ| = |ZX| = √10

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9. A ball is projected with an initial velocity vector $v_0 = 10i + 20j$. The horizontal component is in the x-direction and gravity is $g = 0i - 10j$. Find the maximum height and horizontal range of the ball.

Horizontal velocity is 10 m/s and vertical velocity is 20 m/s

Vertical motion of the ball: $y = v_y t - \frac{1}{2}gt^2$

The ball lands when $y = 0$

$20t - 5t^2 = 0$

$5t(4 - t) = 0$

$t = 0$ (start) or $t = 4$ s (landing)

Maximum height is reached at $t = 2$ s

$h = 20(2) - 5(2)^2 = 40 - 20 = 20$ m

Horizontal motion of the ball: $x = v_x t$

Range $= 10 \times 4 = 40$ m

Answer: Maximum height = 20 m, horizontal range = 40 m

10. A car enters a loop with velocity vector $v = 30j$ and exits with velocity $v' = 30i$. What is the change in velocity vector?

Change $= v' - v = 30\bar{i} - 30\bar{j}$

$|\text{Change}| = \sqrt{((30)^2 + (-30)^2)} = \sqrt{1800} = 30\sqrt{2}$

Answer: Change in velocity = 30i − 30j, magnitude ≈ 42.4 m/s

11. An aeroplane has airspeed $v_p = 20i$ and there is a crosswind $v_w = 50j$. Find the resultant velocity and its magnitude.

Resultant $= v_p + v_w = 20\bar{i} + 50\bar{j}$

$|\text{Resultant}| = \sqrt{((20)^2 + (50)^2)} = \sqrt{(400 + 2500)}$

$= \sqrt{2900} = 10\sqrt{29}$

Answer: Resultant velocity = 20i + 50j, magnitude = $10\sqrt{29}$ ≈ 53.9 km/h

Remember — vectors in the plane

Translation by $ai + bj$: $(x, y) \to (x + a, y + b)$.

$|AB| = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

Parallelogram: AB = DC (equal and parallel).

Projectile: max height $= v_y^2 / 2g$, time $= 2v_y / g$, range $= v_x \times$ time.